Foreword

Mr. Gary Rubinstein has been teaching mathematics at Stuyvesant for 20 years. He is also a dedicated math author, having published "Descartes' Method for Constructing Roots of Polynomials with 'Simple' Curves" in the Convergence, Mathematical Association of America's online journal. In addition, he has written the Barron's Regents Exams and Answers for Algebra I and Algebra II.

It's a great time to be a math student. Back when I was in high school in the late 1980s, there were only two sources to learn from: the teacher and the textbook. I remember my own math teachers very well. They were hard workers and they had a concrete goal that we would ace the Regents exams, but it didn't leave a lot of time for more inspirational and intriguing mathematics. Had there been the Internet and YouTube and access to math journals, I wonder if I would have utilized these great resources.

One of the courses I teach is called Math Research. Over the years, it has become mainly an elective that some freshmen take when they want extra math but don't want to compete on the math team. The course culminates in students writing a research paper in which they learn some interesting math topic and try to extend that topic a little with their own original ideas. Over the years, while grading these papers, I have found myself doing my own research on the topics the students chose. To grade accurately, I needed to fully understand the topic. In these studies I did as part of my grading process, I have learned some of the most interesting mathematics I have ever known.

I once had a student who wanted to write about the math of Astronomy. As modern Astronomy requires a lot of Calculus, I told this student to go back to ancient Astronomy. It turned out that many cultures throughout history have pondered the mystery of the glowing orbs in the night sky. This led to the invention of Trigonometry and, more specifically, spherical Trigonometry. I had no idea that such a thing existed: trigonometry where the sides of the triangles were on the surface of a sphere, and the triangle was a bit like the base of a pyramid where the vertex of the pyramid was at the center of the sphere.

I have some books about spherical Trigonometry that I pick up and read from time to time. There aren't many books published about it nowadays since it is no longer a hot topic. But it is fun to peer into the past and to learn something that few people alive today, even among people who love math, are interested in. It's like being in a world where Shakespeare was not a household name, and I discover an obscure British playwright and dive into these plays, thinking what a shame it is that he is not well known.

There's too much math to master it all. But for any topic, if you desire, you can dive into it and see where it leads. In fact, that might lead you to even more topics and mathematical connections.

And that's what I think the purpose of the *Math Survey* is. It is an opportunity for students writers to follow a mathematical interest and see where it leads. And hopefully the students who read the *Math Survey* are inspired to do a little of their own research on the topics and to follow the path that interests them.

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